

09 Iris Logistic Regression

Markdown Cell 1

Experiment 9 — Logistic Regression using Iris Dataset
Source: IRIS DATASET PDF

Implementation of Logistic Regression using the Iris Dataset from scikit-learn. The PDF also contains a Linear Regression comparison and two scatter plots.

Code Cell 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.linear_model import LinearRegression

iris = load_iris()
```

Code Cell 3

```
dir(iris)
```

Code Cell 4

```
iris.DESCR
```

Code Cell 5

```
iris.target
```

Code Cell 6

```
iris.data
```

Code Cell 7

```
iris.feature_names
```

Code Cell 8

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df.head()
df['target'] = iris.target
df.head()
```

Code Cell 9

```
iris.target_names
```

Code Cell 10

```
df['flower_name'] = df.target.apply(lambda x: iris.target_names[x])
df.tail()
```

Code Cell 11

```
df['flower_name'] = df.target.apply(lambda x: iris.target_names[x])
df.head()
```

Code Cell 12

```
from sklearn.model_selection import train_test_split

x = df.drop(['target', 'flower_name'], axis='columns')
x.head()
```

Code Cell 13

```
y = df.target

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.25, shuffle=True
)
```

Markdown Cell 14

Linear Regression

Code Cell 15

```
model = LinearRegression()  
model.fit(x_train, y_train)  
model.score(x_test, y_test)
```

Markdown Cell 16

Logistic Regression

Code Cell 17

```
model = LogisticRegression()  
model.fit(x_train, y_train)  
model.score(x_test, y_test)
```

Code Cell 18

```
plt.xlabel('sepal length (cm)')  
plt.ylabel('sepal width (cm)')  
plt.scatter(df['sepal length (cm)'], df['sepal width (cm)'], marker='+', color='blue')  
plt.scatter(df['petal length (cm)'], df['petal width (cm)'], marker='*', color='green')  
plt.show()
```